2.11.1 Problems

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4				
2	2				
1	3				
–1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum(x_i-\bar{x}) =$	$\sum(x_i-\bar{x})^2 =$	$\sum(y_i-\bar{y}) =$	$\sum(x_i-\bar{x})(y_i-\bar{y}) =$

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i-\bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i-\bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4				
2	2				
1	3				
–1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i\hat{e}_i =$

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.

f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i/N$.

i. Compute $\hat{\sigma}^2$.

j. Compute $\widehat{\text{var}}(b_2|\mathbf{x})$ and $\text{se}(b_2)$.

a)

$\bar{x} = 1$ $\bar{y} = 2$

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
–1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$	$\sum y_i =$	$\sum(x_i-\bar{x}) =$	$\sum(x_i-\bar{x})^2 =$	$\sum(y_i-\bar{y}) =$	$\sum(x_i-\bar{x})(y_i-\bar{y}) =$
5	10	0	10	0	8

b. $b_2 = \dfrac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} = \dfrac{8}{10} = 0.8$

$\Rightarrow$ one unit increase in x, increase y by 0.8 unit on average.

$b_1 = 2 - 0.8 \times 1 = 1.2$

$\Rightarrow$ $x=0$, the predicted value of y is 1.2.

c. $\sum_{i=1}^{5} x_i^2 = 9+4+1+1+0$
$= 15$ #

$\sum_{i=1}^{5} x_i y_i = 12+4+3+(-1)+0$
$= 18$

$\sum(x_i-\bar{x})^2 = 10 = \sum x_i^2 - N\bar{x}^2 = 15-5 = 10$

$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8 = \sum x_i y_i - N\bar{x}\bar{y} = 18 - 5 \cdot 1.2 = 8$

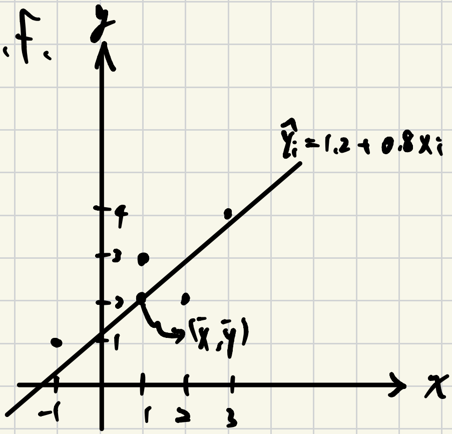
d. $$s_y^2 = \frac{\sum_{i=1}^{5}(y_i-2)^2}{5-1} = \frac{(2^2+0+1^2+(-1)^2+(-2)^2)}{4} = 25$$

$$s_x^2 = \frac{\sum_{i=1}^{5}(x_i-1)^2}{5-1} = \frac{2^2+1^2+0+(-2)^2+(-1)^2}{4} = 2.5$$

$$s_{xy} = \frac{\sum_{i=1}^{5}(y_i-2)(x_i-1)}{5-1} = \frac{2\times2+0+0+(-1)(-2)+(-2)(-1)}{4}$$

$$= 2$$

e.f.



g. $\bar{y} = 2 = b_1 + b_2\bar{x} = 1.2 + 0.8\cdot 1 = 2$

h. $\bar{\hat{y}} = \bar{y} = 2 = \frac{\sum \hat{y}_i}{N} = \frac{10}{5} = 2$

i. $\hat{\sigma}^2 = \frac{\sum(y_i-\hat{y}_i)^2}{N-2} = \frac{\sum \hat{e}_i^2}{5-2} = \frac{3.6}{3} = 1.2$

j. $\widehat{var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum(x_i-\bar{x})^2} = \frac{1.2}{10} = 0.12$

$SE(b_2) = \sqrt{0.12} = 0.3464$

2.22

a.

小班制對總分的影響

	Dependent variable:
	Total Score
Small Class	12.175**
	(5.369)
Constant	916.442***
	(3.675)
Observations	775
R^2	0.007
Adjusted R^2	0.005
Residual Std. Error	74.587 (df = 773)
F Statistic	5.142** (df = 1; 773)
Note:	*p**p***p<0.01

Since b2 = 12.175 > 0, we can say that student from small classes perform better than regular classes on average.

b.

The Effect of Small Class Size on Reading Scores

	Dependent variable:
	Read Score
Small Class	6.924***
	(2.174)
Constant	432.665***
	(1.488)
Observations	775
R^2	0.013
Adjusted R^2	0.012
Residual Std. Error	30.204 (df = 773)
F Statistic	10.142*** (df = 1; 773)
Note:	*p**p***p<0.01

The Effect of Small Class Size on Math Scores

	Dependent variable:
	Math Score
Small Class	5.251
	(3.578)
Constant	483.777***
	(2.449)
Observations	775
R^2	0.003
Adjusted R^2	0.001
Residual Std. Error	49.708 (df = 773)
F Statistic	2.153 (df = 1; 773)
Note:	*p**p***p<0.01

Since 6.924 > 5.251, small class sizes have a stronger impact on reading.

c.

The Effect of Aide on Total Scores

	Dependent variable:
	Total Score
Aide	4.306
	(4.994)
Constant	916.442***
	(3.559)
Observations	837
R^2	0.001
Adjusted R^2	-0.0003
Residual Std. Error	72.232 (df = 835)
F Statistic	0.744 (df = 1; 835)
Note:	*p**p***p<0.01

Since b2 = 4.306 > 0, we can say that student from regular classes with a teacher aide perform better than those without a teacher aide.

d.

The Effect of Aide on Reading Scores

	Dependent variable:
	Reading Score
Aide	2.871
	(2.049)
Constant	432.665***
	(1.460)
Observations	837
R^2	0.002
Adjusted R^2	0.001
Residual Std. Error	29.635 (df = 835)
F Statistic	1.964 (df = 1; 835)
Note:	*p**p***p<0.01

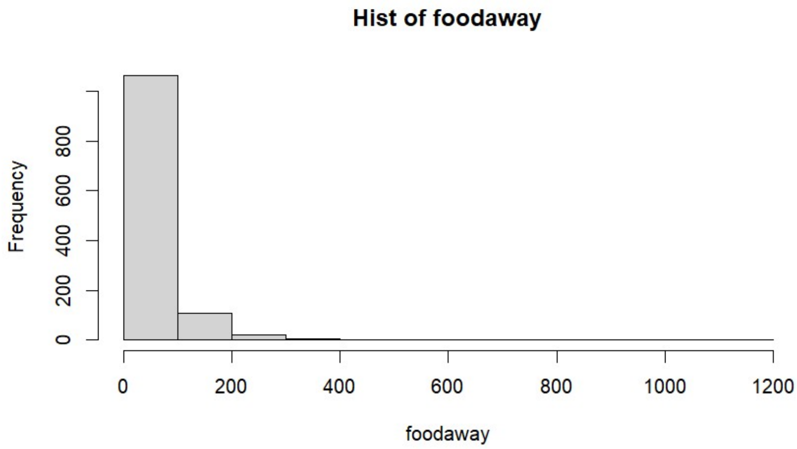
The Effect of Aide on Math Scores

	Dependent variable:
	Math Score
Aide	1.435
	(3.304)
Constant	483.777***
	(2.355)
Observations	837
R^2	0.0002
Adjusted R^2	-0.001
Residual Std. Error	47.792 (df = 835)
F Statistic	0.189 (df = 1; 835)
Note:	*p**p***p<0.01

Since 2.871 > 1.435, classes with a teacher aide have a stronger positive impact on reading.

2.25

a.




```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00   12.04   32.55   49.27   67.50 1179.00
```

b.

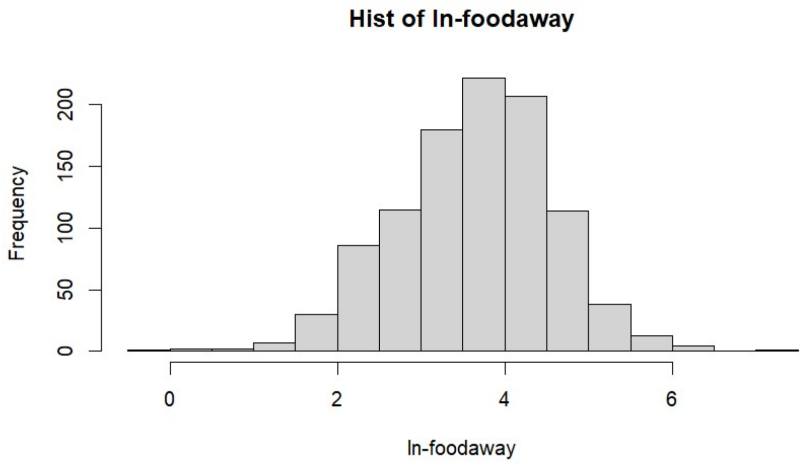
```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00   21.67   48.15   73.15   90.00 1179.00
```

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00   14.44   36.11   48.60   68.67  416.11
```

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00    9.63   26.02   39.01   52.65  437.78
```

Households with higher education levels tend to spend more on food away from home.

c.



FOODAWAY and ln(FOODAWAY) have different numbers of observations because the minimum value of FOODAWAY is 0.00. Since the natural log of zero is undefined, those observations are excluded from the log-transformed data.

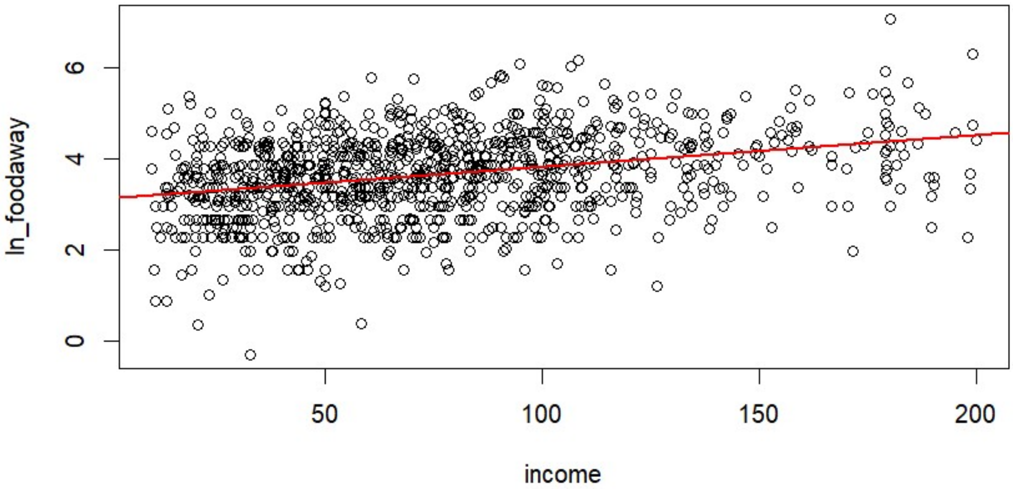
d.

For every one unit of INCOME increase in monthly income, the expenditure on food away from home is expected to change by approximately 0.007 percent.

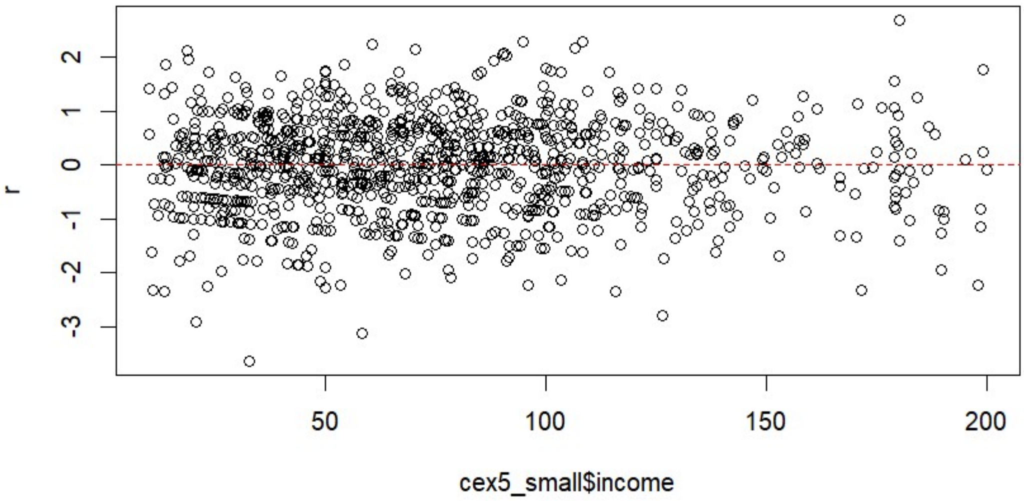
The Effect of Foodaway on Income

	Dependent variable:
	Ln Foodaway
Income	0.007^{***}
	(0.001)
Constant	3.129^{***}
	(0.057)
Observations	1,022
R^2	0.098
Adjusted R^2	0.097
Residual Std. Error	0.876 (df = 1020)
F Statistic	111.150^{***} (df = 1; 1020)
Note:	*p**p***p<0.01

e.



f.



The residuals are scattered around the zero line. However, the spread seems to increase slightly as income increases, which might suggest heteroskedasticity.